PALEOMAGNETIC STUDIES OF THE EARLY CARBONIFEROUS ST. JOE LIMESTONE, ARKANSAS

G. R. Scott[1]

Geoscience Program, University of Texas at Dallas, Richardson, Texas 75080

Abstract. The Kinderhookian (early Tournaisian) St. Joe Limestone was sampled at two quarries in the Ozark Plateau. A total of 110 oriented core samples were collected from brick red to gray crinoidal limestone and mudstone at approximately 20-cm stratigraphic intervals. All samples were reversely magnetized. A selection procedure based upon directional stability of demagnetized vectors restricts the stable data population to 39 and 43 samples at each site with paleomagnetic pole positions of 46.2°W longitude, 37.0°S latitude, α_{95} = 1.67, κ = 189; and 50.7°W longitude, 40.6°S latitude, α_{95} = 1.48, κ = 218, respectively. These results from 'cratonic' North America are the first from early Carboniferous rocks and delimit a period of active apparent polar wander to the latest Devonian and earliest Carboniferous. The relative motion involved a clockwise rotation of the plate. This was immediately followed by a quasi-static period during the Carboniferous. However, beginning in the latest Carboniferous the North American plate rotated counterclockwise with a large northerly component of motion about a tectonic rotational point at 62°N, 150°E.

Introduction

The early Carboniferous St. Joe Limestone is exposed throughout the central Ozark Plateau in Arkansas, Missouri, and Oklahoma and generally consists of 5-30 m of brick red, red, green, and gray crinoidal limestone. The red color is the result of a hematitic mud matrix which is interstitial to the medium-to-coarsely disarticulated crinoid columnals and ossicles. Samples for paleomagnetic analysis were collected from two quarries in north central Arkansas: at St. Joe, 34 km southeast of Harrison; and near Jasper, 23 km south of Harrison (Figure 1).

The primary objective was to provide a sampling of the paleomagnetic field during the early Carboniferous for cratonic North America. Although no paleomagnetic results have been reported for this particular age, Knowles and Opdyke [1968] have studied a middle Carboniferous unit, the Mauch Chunk Formation in Pennsylvania. It was anticipated that paleomagnetic results from the present study would provide additional knowledge of the North American apparent polar wander path and would aid in the plate tectonic reconstructions for the late Paleozoic era.

Geology

The unit referred to in this paper as the St. Joe Limestone [Thompson and Fellows, 1970] has been proposed as the St. Joe Group in Oklahoma and Missouri [Huffman et al., 1966] and as the St. Joe member of the Boone Formation in Arkansas. This unit is overlain by a gray cherty limestone unit named the Boone Formation in Arkansas. Conformably below the St. Joe Limestone is the Bachelor Formation, a thin (15-20 cm) gray to light green conglomeratic sandstone. However, the Bachelor Formation and at least locally the Sylamore Sandstone (Figure 2) is regionally suprajacent with a major erosional and angular unconformity formed upon various stratigraphic units from Ordovician to Devonian in age [Freeman and Schumacker, 1969].

The 'type' locality for the St. Joe Limestone is at a quarry 2.5 km west of the small town of St. Joe, Arkansas [Thompson and Fellows, 1970]. This same site was used for the present study. The term type locality is used to denote the principal reference locality, since no actual type section was designated by early workers.

Age

Thompson and Fellows [1970] used conodonts to establish eight biostratigraphic zones or subzones at the St. Joe quarry, in the St. Joe Limestone, and in the Bachelor Formation. These biostratigraphic zones were assigned principally to the upper Kinderhookian stage and subordinately to the lower Osagean stage. The conodont assemblage can also be compared with those of Europe and Australia, where a similar assemblage is pre-

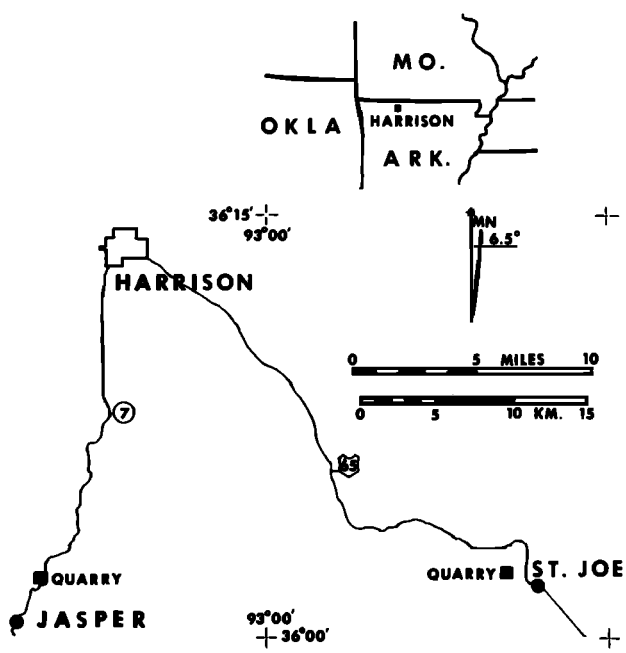


Fig. 1. Generalized locality map showing general features and access to quarries sampled near St. Joe and Jasper, Arkansas.

[1]Present address: Geophysics Laboratory, Marine Science Institute, The University of Texas at Austin, Galveston, Texas 77550

Copyright 1979 by the American Geophysical Union.

Paper number 9B0734.
0148-0227/79/009B-0734$01.00

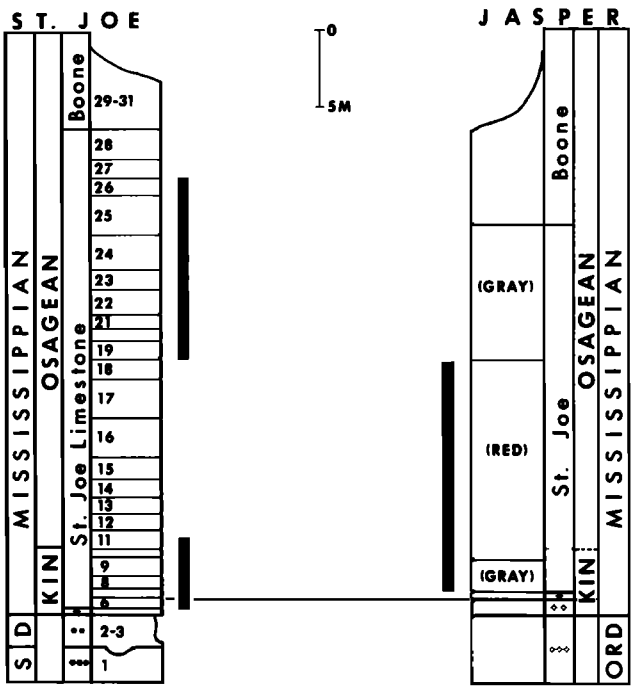


Fig. 2. Stratigraphy and age of sections near St. Joe and Jasper, Arkansas. Detailed stratigraphy at St. Joe quarry after Thompson and Fellows [1970]. Heavy bar shows stratigraphic interval over which paleomagnetic samples were collected. Single star, Bachelor formation; double star, Chattanooga Shale; triple star, St. Clair Limestone; double crossed circles, Sylamore Sandstone; triple crossed circles, Fernvale Limestone. Correspondence based on conodont zone Siphonodella quadruplicata--S. crenulata (Jasper, from Freeman and Schumacher [1969] to S. lobata--S. crenulata zone (St. Joe, from Thompson and Fellows [1970]).

sent in the middle and upper Tournaisian stage (lowest Carboniferous) [Jenkins, 1974].

The age of the St. Joe Limestone at the Jasper quarry has not been directly examined. However, Freeman and Schumacker [1969] did establish a

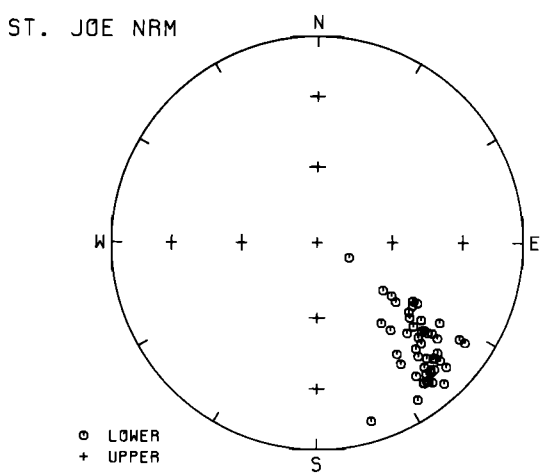


Fig. 3. Natural remanent magnetization directions from the St. Joe quarry. Geographic coordinates, magnetic north; present inclination, +66°.

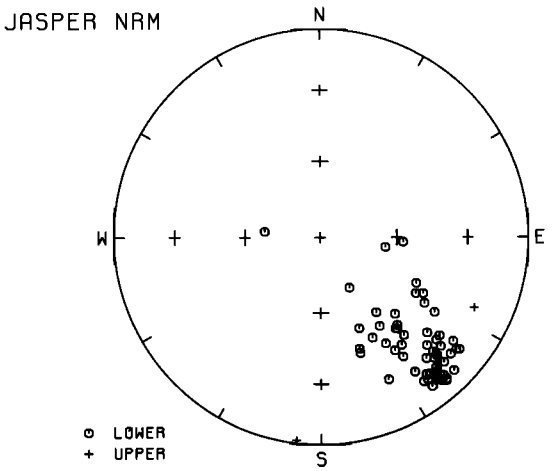


Fig. 4. Natural remanent magnetization directions from the Jasper quarry. Geographic coordinates, magnetic north; present inclination, +66°.

correspondence also using conodonts, to the middle Kinderhookian stage for the Sylamore Sandstone underlying the St. Joe Limestone at Jasper, Arkansas. This supports a general assumption that the rocks in the two sections sampled in this study—30 km apart—are broadly correlative in age, in addition to their being lithostratigraphic equivalents.

Lithology and Sampling

Lithologically, the St. Joe Limestone is similar at both quarries and can be generally characterized as a silty bioclastic micrite. The

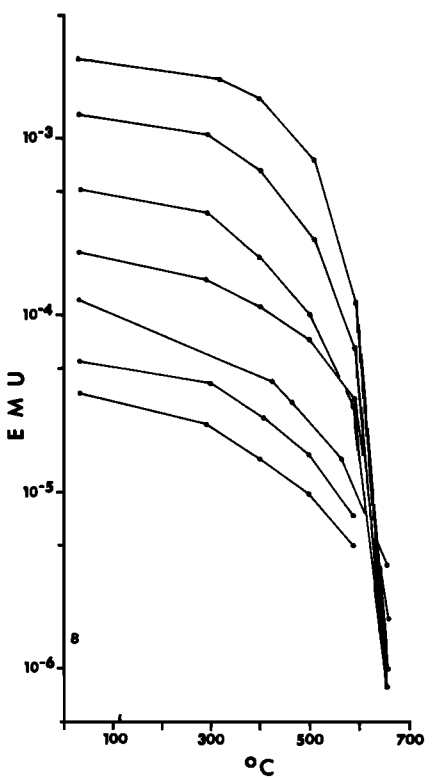


Fig. 5. Thermal demagnetization curves for representative samples from both quarries. Remanence is total moment for 11.1-cm^3 specimens.

21562202b, 1979, B11, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/JB084iB11p06277 by Curtin University Library, Wiley Online Library on [25/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

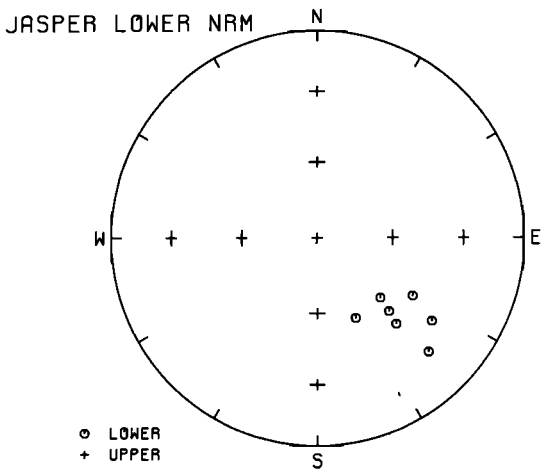


Fig. 6. Natural remanent magnetization directions from lowest 1.4 m of the Jasper quarry. Geographic coordinates, magnetic north.

percentage of reddish silt or mud increases markedly downward in each section but decreases abruptly in the basal units to form a greenish gray bioclastic micrite. A similar greenish gray color is also present in the subjacent Bachelor Formation. At the St. Joe section, gray limestone is characteristic of the upper 2.3 m with the remaining 20.6 m composed of a brick red to pink crinoidal limestone with subordinate red and green mudstone and shale. A measured section at this quarry was given by Thompson and Fellows [1970, pp. 192-196]. In the present study, 49 oriented samples were collected across their section.

The St. Joe Limestone near Jasper is 17.2 m thick. The upper 6.4 m are composed of thickly bedded, gray, crinoidal micrite. The next 9.4 m are composed of medium to thinly bedded, brick red to pink, crinoidal limestone with brick red shale prominent in the lower part. The lowest 1.4 m are composed of light gray to brownish gray pyritic bioclastic limestone. Sixty-one oriented samples were collected at this quarry located 3 km north of Jasper, Arkansas, on State Route 7.

Cyclindrical cores, 2.5 cm in diameter, 4-10 cm in length, were oriented in situ with an inclinometer and a magnetic compass. Samples were drilled at measured stratigraphic intervals of approximately 20 cm, using a portable gasoline-powered drill. Sampling was concentrated in, but not restricted to, the brick red colored lithologic units.

Magnetic Properties and Laboratory Analysis

The bottommost specimen (2.2-cm length, 11.1-cm^3 volume) of each core sample was placed in a low field area during the laboratory analysis phase of this study. This included a minimum 10^7-s time period prior to the measurement of the natural remanent magnetization (NRM). The low field area was a 3.3 x 3.3 x 3.3 m doubly shielded μ metal room, designed after Patton and Fitch [1962] with a maximum field in the interior of 200 gammas. In addition, two cylindrical shields were utilized during the cooling phase of thermal demagnetization, which reduced the residual magnetic field to no greater than 5 γ. Alternating field (af) demagnetization was carried out by a 400-Hz decaying field with a two-axis tumbler. The residual magnetic field in which the af demagnetization took place was that of the room (200 γ).

Measurement of the remanent magnetization was made on a two-axis cryogenic magnetometer (SCT 1) with an approximate resolution of 5 x 10^{-9} emu (total moment). For magnetic moments below 1 x 10^{-6} emu (total moment), three or more consecutive measurements were made, with the resultant vector sum being used as the magnetization direction and the resultant scalar divided by the number of measurements taken as the intensity of magnetization. This procedure was initiated in response to apparently random short-term drift of intensity and direction observed in some lower intensity specimens. Such deviations were not an artifact of the magnetometer, since low-intensity calibration samples and other low-intensity samples from this study did not reveal this drift behavior. It appears to be a property of only certain samples.

The NRM directions of samples from the St. Joe Limestone are shown in Figures 3 and 4. The NRM populations at both sites display a streaked or smeared distribution between a characteristic reversed late Paleozoic direction (subhorizontal and toward the southeast) and the present or recent magnetic field direction (present field declination, 6.5°; inclination, +66°). The contribution of the secondary disturbing field direction is generally minor, with only 5% of the sample directions lying within 50° of the Quaternary magnetic direction. This type of NRM population distribution indicates moderate magnetic stability versus time. The surface weathering common in this region was largely avoided by sampling in quarries, minimizing the NRM component aligned with the recent geomagnetic field.

Specimens from the St. Joe quarry were subjected to partial thermal demagnetization at temperatures of 300°, 400°, 500° and 590°C and a few at 655°C (Figure 5). The higher-temperature demagnetizing steps were deleted on those

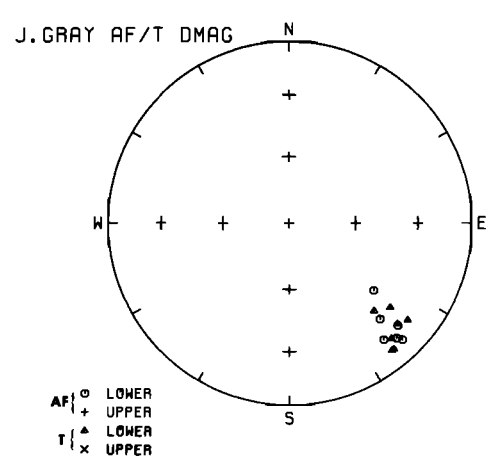


Fig. 7. Directions from gray sulfide-rich samples from lowest 1.4 m of Jasper quarry displaying maximum migration toward characteristic endpoint of the streak, prior to apparently random direction dispersal. Magnetic north, no structural correction has been applied.

21562202b, 1979, B11, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/JB084iB11p06277 by Curtin University Library, Wiley Online Library on [25/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

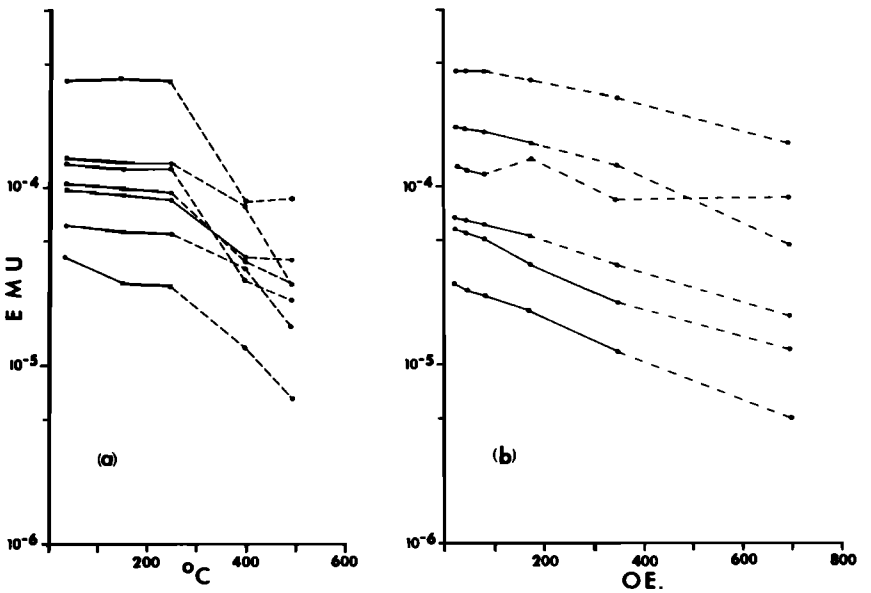


Fig. 8. (a) Thermal demagnetization curves for nonred samples from Jasper quarry. (b) Alternating field demagnetization curves for nonred samples from Jasper quarry. Same samples as in Figure 8a but from adjacent specimen. (Dashed lines lead to points with erratic directional data.)

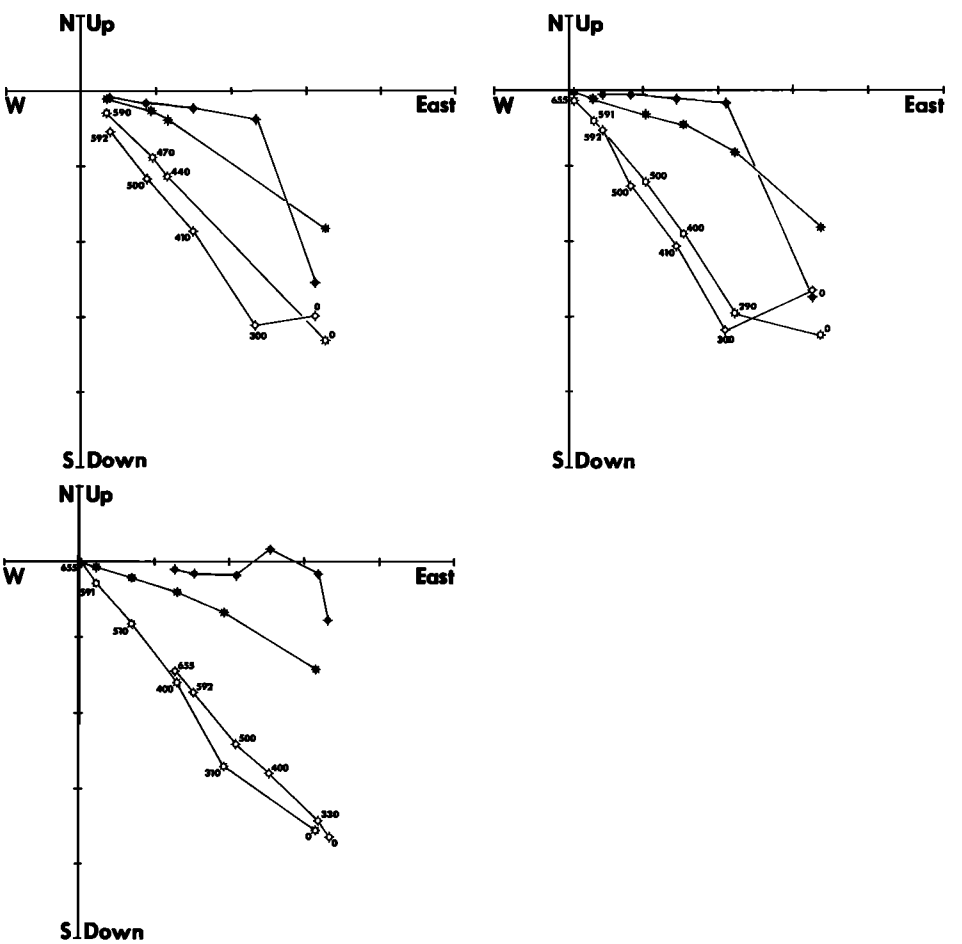


Fig. 9. Orthogonal progressive demagnetization diagrams for stable samples from St. Joe and Jasper quarries displaying vector endpoints after partial thermal demagnetization. Two samples are shown in each diagram; numbers beside vector endpoints are in degree centigrade. Open symbols are in horizontal plate (delination); solid symbols are in vertical plane (projection of inclination along E-W axis). Coordinates are geographic, magnetic north.

21562202b, 1979, B11, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/JB084iB11p06277 by Curtin University Library, Wiley Online Library on [25/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

TABLE 1. Paleomagnetic Results, Stable Vector Trends

Field	D, deg	I, deg	Interval,* cm	T,† °C
		St. Joe Quarry		
MA 200	146.7	0.7	...	330/400 (2)
MA 201	151.9	4.3	18	400/500 (2)
MA 202	151.3	-1.6	23	330/400 (2)
MA 203	143.1	8.6	23	500/592 (2)
MA 204	145.5	6.7	18	500/655 (3)
MA 206	149.6	2.2	54	500/592 (2)
MA 207	152.4	4.1	18	300/410 (2)
MA 208	150.3	4.8	23	300/410 (2)
MA 209	152.3	2.0	30	300/410 (2)
MA 210	152.6	1.3	25	300/410 (2)
MA 211	153.5	4.0	28	300/410 (2)
MA 212	152.2	5.4	30	300/592 (4)
MA 213	146.9	6.5	23	410/592 (3)
MA 214	148.1	9.1	36	410/592 (3)
MA 215	145.7	4.4	28	300/592 (4)
MA 216	146.4	7.7	30	410/592 (3)
MA 217	148.6	18.1	25	300/410 (2)
MA 219	151.7	13.3	63	500/592 (2)
MA 220	153.7	19.4	23	410/500 (2)
MA 221	143.9	12.9	36	410/592 (3)
MA 222	149.3	8.3	15	410/592 (3)
MA 223	150.4	14.7	13	410/500 (2)
MA 224	145.7	1.9	28	300/300 (3)
MA 225	150.1	5.3	23	410/500 (2)
MA 227	150.1	10.3	59	290/591 (4)
MA 228	149.2	3.7	33	400/500 (2)
MA 229	149.3	4.4	33	400/591 (3)
MA 230	150.8	8.0	10	290/591 (3)
MA 232	149.4	8.0	64	400/500 (2)
MA 233	150.3	12.1	33	290/400 (2)
MA 240	151.0	4.4	870	400/500 (2)
MA 241	150.7	9.2	25	400/591 (3)
MA 242	149.0	10.4	20	400/500 (2)
MA 243	146.4	13.2	36	400/655 (4)
MA 244	147.4	13.6	30	400/591 (3)
MA 245	150.0	8.8	23	400/500 (2)
MA 246	146.6	12.9	25	400/591 (3)
MA 247	147.8	13.2	15	510/591 (2)
MA 254	148.5	12.5	23	400/510 (2)
MA 249	146.6	13.6	28	400/591 (3)
MA 250	150.1	10.8	20	400/510 (2)
MA 251	150.0	11.1	30	400/510 (2)
MA 253	155.5	10.2	33	310/510 (3)
Average	149.3	8.2		
α_{95} = 1.48				
κ = 218				
N = 43				
		Jasper Quarry		
MA 051	149.0	9.2	...	427/590 (3)
MA 052	144.8	5.2	13	471/563 (2)
MA 053	144.3	6.5	13	427/563 (3)
MA 054	144.8	10.9	10	427/563 (3)
MA 055	136.8	23.7	13	471/563 (2)
MA 056	141.7	12.2	15	427/471 (2)
MA 057	142.3	12.7	15	427/563 (3)
MA 059	150.5	13.0	41	427/471 (2)
MA 060	141.9	4.4	18	427/471 (2)
MA 063	149.0	13.8	60	427/471 (2)
MA 064	144.6	5.2	25	471/563 (2)
MA 065	145.5	8.3	18	427/471 (2)
MA 066	149.6	5.4	18	427/510 (2)
MA 067	148.3	6.7	30	427/590 (3)
MA 068	150.8	5.7	30	427/510 (2)
MA 070	152.6	6.7	61	427/500 (2)

TABLE 1. (continued)

Field	D, deg	I, deg	Interval,* cm	T,† °C
		Jasper Quarry		
MA 074	134.3	12.5	90	440/471 (2)
MA 075	150.4	18.5	23	440/471 (2)
MA 076	141.3	10.0	18	440/471 (2)
MA 081	146.1	13.1	84	440/590 (3)
MA 083	140.3	14.2	31	440/590 (3)
MA 084	135.3	9.1	18	440/590 (3)
MA 085	134.8	10.0	15	440/590 (5)
MA 086	134.1	11.0	18	440/590 (3)
MA 087	142.3	9.4	15	471/590 (2)
MA 088	144.8	6.9	13	470/590 (2)
MA 089	144.1	12.7	13	420/590 (3)
MA 091	147.0	7.9	41	470/590 (2)
MA 092	144.3	8.6	15	470/590 (2)
MA 093	147.6	6.9	10	420/470 (2)
MA 094	144.4	9.8	15	470/590 (2)
MA 095	144.7	9.3	15	420/590 (3)
MA 096	144.3	8.9	15	420/590 (3)
MA 097	146.1	10.1	8	420/470 (2)
MA 098	144.7	8.4	10	420/590 (3)
MA 100	143.5	9.6	38	420/590 (3)
MA 101	143.0	7.8	10	471/590 (2)
MA 102	146.5	10.9	13	420/592 (3)
MA 103	139.6	13.0	23	420/471 (2)
Average	144.1	10.0		
α_{95} = 1.67				
κ = 189				
N = 39				

D = Declination
I = Inclination
T = Temperature
*Interval is the stratigraphic distance below the previous sample.
†Number in parentheses is number of stable demagnetized steps.

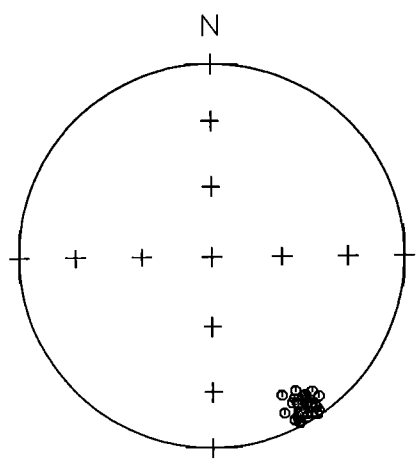


Fig. 10. Selected stable demagnetized directions for samples from the St. Joe quarry. Data are structurally corrected (strike, 232°; dip, 3°NW) and referenced to true north. Mean declination, 149.3°; mean inclination, 8.2°; N = 43; α_{95} = 1.48°; κ = 218; R = 42.8075.

21562202b, 1979, B11, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/JB084iB11p06277 by Curtin University Library, Wiley Online Library on [25/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

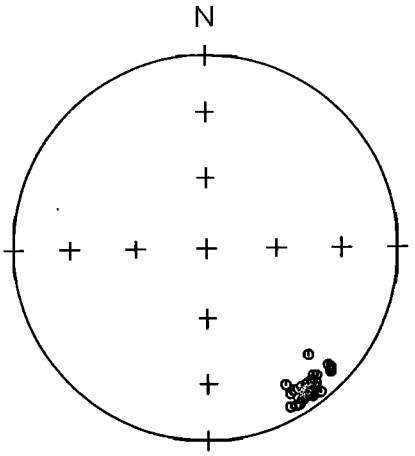


Fig. 11. Selected stable demagnetized directions for samples from the Jasper quarry. Data are structurally corrected (strike, 227°; dip, 2°NW) and referenced to true north. Mean declination, 144.1°; inclination, 10.0°; N = 39; α_{95} = 1.67°; κ = 189; R = 38.7989.

specimens whose remanence stability range had been previously exceeded. A similar procedure was applied to the specimens from the quarry near Jasper, with temperatures of approximately 425°, 480°, 570°, and 655°C (Figure 5).

Samples from the lowest 1.4 m of the Jasper section are gray sulfide-rich limestones. The NRM directions of these samples (Figure 6) are smeared or streaked toward the present magnetic field direction, slightly more so than the red colored samples (Figure 4). Thermal demagnetization of 140°, 240°, 400°, and 490°C (Figure 7) revealed a directional trend toward the primary end of the streak up to 240° or 400°C and then a dispersal of directions by 490°C. Alternating field demagnetization was applied to a second specimen of these nonred samples using maximum fields of 22, 44, 88, 175, 350, and 700 Oe. The partially demagnetized directions migrated down the streak toward the primary end, with increasing field strength until 175 Oe but no higher than 350 Oe (Figure 7). The directions dispersed to apparently random directions with the magnetic intensities still in the range of 0.39 x 10^{-4} to 0.50 x 10^{-6} emu total moment (Figure 8).

Analysis of Remanence

Of highest priority in research such as this, which seeks a paleomagnetic pole position, is to restrict the sample population to data that demonstrate stable magnetic properties arising from the deposition, lithification, or diagenesis of these sedimentary rocks. Those sample directions that contain a detectable secondary component of magnetization should be rejected from the population used for calculating a paleomagnetic pole position. This conceptual restriction is not easily applied to practical situations. The orthogonal demagnetization diagram [Zijderveld, 1967] is useful in visualizing the presence of

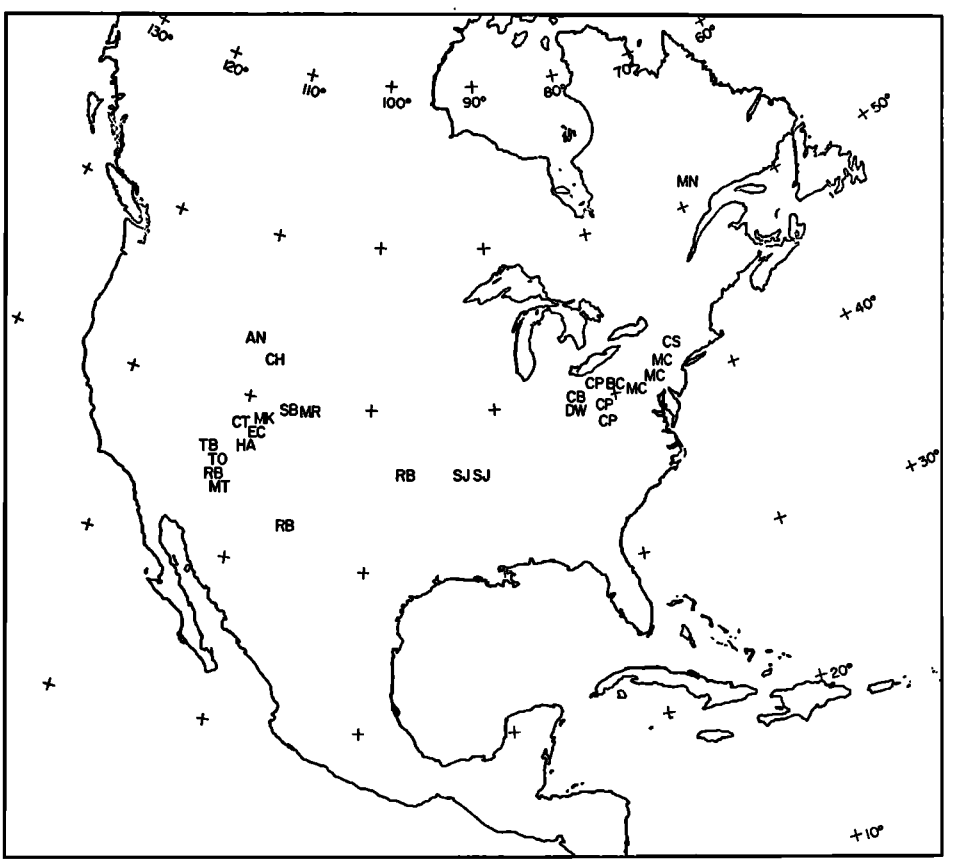


Fig. 12. Sampling sites for Devonian through early Triassic units. See Table 2 for specific site labels.

21562202b, 1979, B11, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/JB084iB11p06277 by Curtin University Library, Wiley Online Library on [25/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

TABLE 2. Paleomagnetic Pole Positions--North American Cratonic Sites.

Symbol	Rock Unit	Pole Position	Reference*
		Triassic (Early)	
AN	Ankareh	47°N, 103°E	Grubbs and Van der Voo [1976]
CH	Chugwater	53°N, 104°E	Grubbs and Van der Voo [1976]
MK	Moenkopi	57°N, 100°E	14.299
MN	Manicougen	58°N, 88°E	9.51, 9.52
SB	State Bridge	56°N, 100°E	Christensen [1974]
TE	Mean	54°N, 99°E	
		Permian	
RB	Red beds	48°N, 114°E	13.47
TO	Toroweap	47°N, 103°E	11.65
P	Mean	48°N, 108°E	
		Permian (Early) to Carboniferous (Late)	
BC	Brush Creek	35°N, 125°E	Payne et al. [1978]
CT	Cutler	44°N, 119°E	13.50, 13.51
EC	Elephant Canyon	44°N, 120°E	13.52
HA	Halgaito	49°N, 117°E	11.67
MR	Maroon	41°N, 124°E	Christensen [1974]
P/C	Mean	42°N, 121°E	
		Carboniferous (Pennsylvanian)	
CP	Coal Pole	37°N, 132°E	Kopacz and Noltimier [1976]
		Carboniferous (Mississippian)	
MC	Mauch Chunk	43°N, 127°E	10.119
SJ	St. Joe	39°N, 132°E	this study
CE	Mean	41°N, 130°E	
		Devonian (Late and Middle)	
CB	Columbus	45°N, 120°E	14.366
CS	Catskill	47°N, 117°E	Kent and Opdyke [1977]
DW	Delaware	48°N, 118°E	14.365
MT	Martin	56°N, 109°E	Elston and Bressler [1977]
TB	Temple Butte	53°N, 115°E	Elston and Bressler [1977]
D	Mean	50°N, 116°E	

* Numbered references are from McElhinny's lists in the Geophysical Journal 1968-1978, 'Paleomagnetic Directions and Pole Positions.'

multicomponent magnetization. Vector endpoints from partial thermal demagnetization are projected onto two orthogonal planes. When these endpoints display a trend toward the origin, a single component of magnetization is present. This means that no rotation of the magnetic vector has occurred at increasing temperatures of demagnetization. However, if a straight line is formed which does not trend to the origin, it indicates at least a two-component remanence with nonoverlapping blocking temperature spectra. Any curved trend of vector endpoints indicates the decay of multiple magnetic components with overlapping blocking spectra.

The problem still exists as to what constitutes a linear trend of vector endpoints. In the absence of any mathematical criteria for selecting linear trends, a subjective examination (hopefully critial) was made of orthogonal diagrams from all samples. Some good examples of linear trends toward the origin can be seen in Figure 9. These selection criteria admitted 43 of 49 samples from the St. Joe site and 39 of 61 samples from the Jasper site.

It is argued that no single vector is more important in these linear trends than another; therefore all vectors are summed to form the average direction of the trend. These data are listed in Table 1 with the average sample directions plotted on Figures 10 and 11. The paleomagnetic parameters and statistical information from each site were then calculated using those samples which passed these selection criteria. The St. Joe quarry site has a structurally corrected mean pole position of 40.6°S, 50.7°W; dp = 0.8; dm = 1.5. The Jasper section has a mean pole position of 37.0°S, 46.2°W; dp = 0.8; dm = 1.7.

Discussion

The following discussion will be restricted to North American sites west of the Appalachian orogenic zone and east of the Cordilleran fold and thrust belt. These will be considered 'cratonic' North American sites (Figure 12). Present geographic coordinates and directions will be used.

This investigation produced results similar to the slightly younger (Visean) pole (43°N, 127°E) from the Mauch Chunk Formation [Knowles and Opdyke, 1968]. The St. Joe poles are also close to a general late Carboniferous pole (37°N,

21562202b, 1979, B11, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/JB084iB11p06277 by Curtin University Library, Wiley Online Library on [25/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

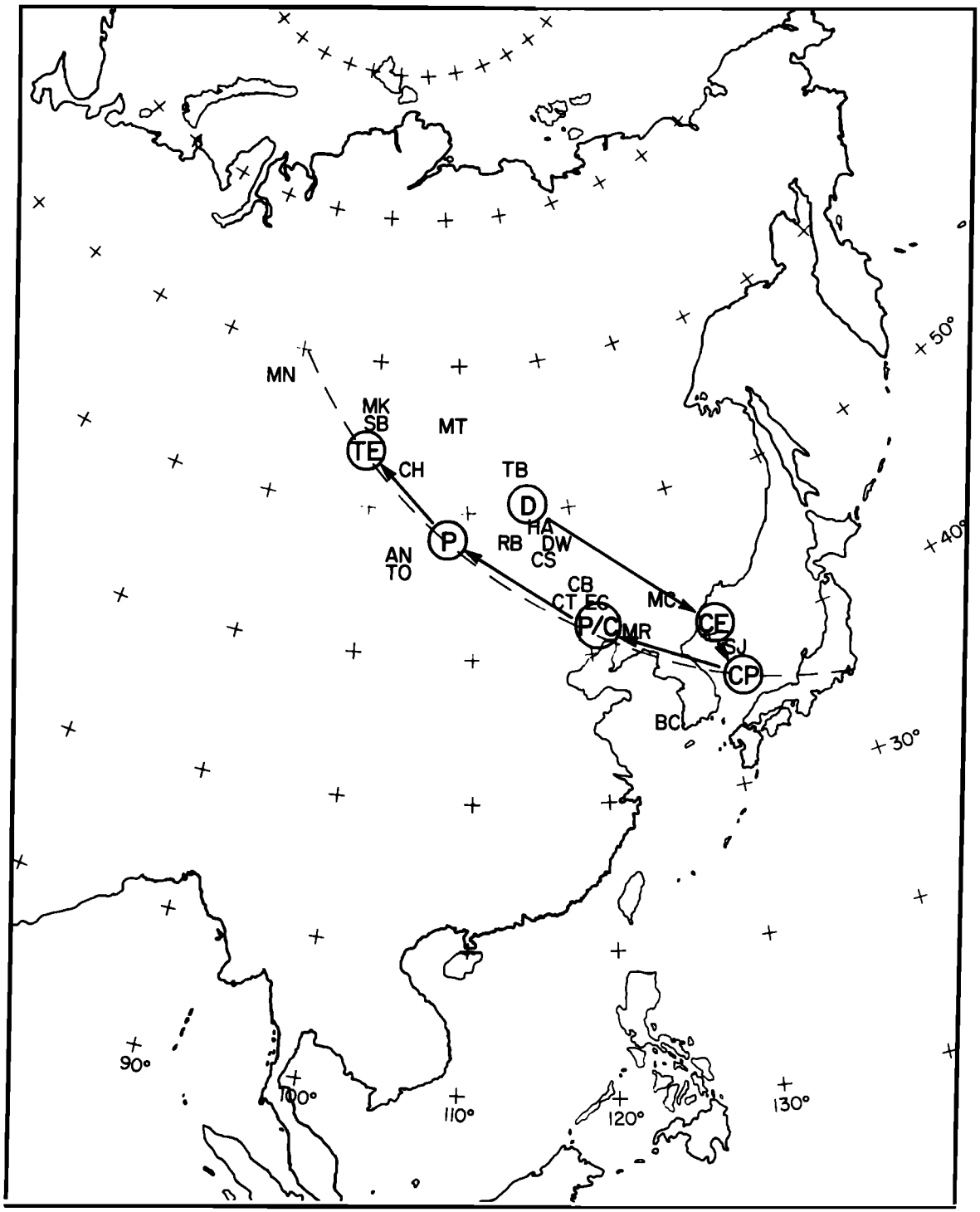


Fig. 13. North pole positions for sites within 'cratonic' North America (from Table 2). Dashed line is the 28° small circle about the tectonic rotational point 62°N, 150°E.

131.6°E) from nine coal and clay units in the eastern United States [Kopacz and Noltimier, 1976]. However, no correspondence can be made with the group of middle and late Devonian poles located at approximately 50°N, 116°E. This indicates a period of active plate motion restricted to the latest Devonian and earliest Carboniferous (345 m.y.), specifically the Famennian-early Tournaisian ages.

A compilation of late Paleozoic results from cratonic North America is given in Table 2 and Figure 13. The late Devonian-early Carboniferous plate motion appears to slow down and reverse direction during the Carboniferous, producing an apparent polar wander path which virtually doubles back on itself, before proceeding along a uniform trend into the early Mesozoic. Such sharp turning points in a path are referred to as hairpins [Irving and Park, 1972] punctuating the paleomagnetic record and directly indicating an

21562202b, 1979, B11, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/JB084iB11p06277 by Curtin University Library, Wiley Online Library on [25/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

alteration in the existing drift direction and velocity. Prior to the hairpin the relative plate motion involved a clockwise rotation about a rotational point close to southwestern North America, producing southerly motion in the eastern part of the continent. During most of the Carboniferous the plate motion diminished greatly (alternatively, the rotational point shifted to coincide with the paleomagnetic pole). The North American plate began a markedly different drift regime by the latest Carboniferous with counterclockwise rotation and a large northerly component of motion. A tectonic rotational point near 62°N, 150°E produced a small circle through all mean poles into the Early Triassic (Figure 13).

Acknowledgment. Research was partially supported by National Science Foundation grant DES 74-1386. C. E. Helsley brought to my attention the presence of hematitic strata in northern Arkansas. T. Beach aided in stratigraphic research. S. C. Reeve and S. K. Parks assisted in sample collection. The Robertson family allowed access to the abandoned St. Joe Quarry. Improvements in the manuscript were suggested by J. Hillhouse. University of Texas at Dallas Geoscience contribution 344.

References

Christensen, F. D., A paleomagnetic investigation of the Permo-Carboniferous Maroon and Upper Permian-Lower Triassic State Bridge Formation in north central Colorado, M.S. thesis, 141 pp., Univ. of Tex. at Dallas, Richardson, 1974.

Elston, D. P., and S. L. Bressler, Paleomagnetic poles and polarity zonation from Cambrian and Devonian strata of Arizona, Earth Planet. Sci. Lett., 36, 423-433, 1977.

Freeman, T., and D. Schumacher, Qualitative pre-Sylamore (Devonian-Mississippian) physiography delinated by onlapping conodont zones, northern Arkansas, Geol. Soc. Amer. Bull., 80, 2327-2334, 1969.

Grubbs, K. L., and R. Van der Voo, Structural deformation of the Idaho-Wyoming overthrust belt (U.S.A.), as determined by Triassic paleomagnetism, Tectonophysics, 33, 321-336, 1976.

Huffman, G. G., J. M. Langton, and J. M. Hancock, Jr., Geology of northern Adair County, Okla. Geol. Surv. Circ., 68., 50 pp., 1966.

Irving, E., and J. K. Park, Hairpins and superintervals, Can. J. Earth Sci., 9, 1318-1324, 1972.

Jenkins, T. B. H., Lower Carboniferous conodont biostratigraphy of New South Wales, Paleontology, 17, 909-924, 1974.

Kent, D. V., and N. D. Opdyke, Paleomagnetism of Devonian red beds from New York State (abstract), Eos Trans. AGU, 58, 375, 1977.

Knowles, R. R., and N. D. Opdyke, Paleomagnetic results from the Mauch Chunk Formation: A test of the origin of curvature in the folded Appalachians of Pennsylvania, J. Geophys. Res., 73, 6515-6526, 1968.

Kopacz, M. A., and H. C. Noltimier, The Coal Pole and the Kittanning Reversal (abstract), Eos Trans. AGU, 57, 238, 1976.

Patton, B. J., and J. L. Fitch, Design of a room-size magnetic shield, J. Geophys. Res., 67, 1117-1121, 1962.

Payne, M., S. Shulik, B. Cain, V. A. Schmidt, J. Donahue, and H. B. Rollins, A Pennsylvanian (Conemaughian) pole position for the Brush Creek limestone from western Pennsylvania, USA (abstract), Eos Trans. AGU, 59, 264, 1978.

Thompson, T. L., and L. D. Fellows, Stratigraphy of Kinderhookian and Osagean rocks of southwestern Missouri and adjacent area, Mo. Geol. Surv. Water Resour. Rep. Invest., 45, 263 pp., 1970.

Zijderveld, J. D. A., AC demagnetization of rocks: Analysis of results, in Methods in Paleomagnetism, edited by D. W. Collinson, K. M. Creer, and S. K. Runcorn, pp. 254-286, Elsevier, New York, 1967.

(Received September 16, 1977;
revised April 27, 1979;
accepted April 30, 1979.)

21562202b, 1979, B11, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/JB084iB11p06277 by Curtin University Library, Wiley Online Library on [25/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License